CLASS 6 — MATHS CHAPTER TEST SET A

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A figure has line symmetry if: 1

(a) it can be folded along a line so that the two halves match
(b) it has angles
(c) it has curved edges
(d) it is coloured

2. How many lines of symmetry does an equilateral triangle have? 1

(a) 6 **(b)** 3 **(c)** 2 **(d)** 1

3. How many lines of symmetry does a regular hexagon have? 1

(a) 6 **(b)** 8 **(c)** 3 **(d)** 4

4. The image of a figure in a mirror is called its: 1

(a) translation **(b)** rotation **(c)** area **(d)** reflection

5. The order of rotational symmetry of a square is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 8

6. The smallest angle by which a regular hexagon can be rotated to coincide with itself is: 1

(a) 30° **(b)** 60° **(c)** 90° **(d)** 120°

7. The letter S has: 1

(a) no symmetry **(b)** line symmetry
(c) rotational symmetry of order 2 **(d)** order 3

8. A rhombus (not a square) has how many lines of symmetry? 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 0

9. The mirror image of the letter ‘b’ in a vertical mirror to its right looks like: 1

(a) d **(b)** b **(c)** p **(d)** q

10. Which has no line of symmetry? 1

(a) Square **(b)** Parallelogram
(c) Circle **(d)** Isosceles trapezium

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is line symmetry? Give an example. 2

12. How many lines of symmetry does a rectangle have? Name them. 2

13. Name three letters with a vertical line of symmetry. 2

14. What is rotational symmetry? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the lines of symmetry of an equilateral triangle, a rectangle and a circle. 3

16. Explain with an example how to check line symmetry by paper folding. 3

17. Write the order and angle of rotational symmetry of a square. 3

OR

Which capital letters have both line and rotational symmetry? Give two. 3

18. Explain the term ‘reflection’ with a diagram idea of the letter F in a mirror. 3

OR

A regular hexagon: find the number of lines of symmetry, the order and the angle of rotational symmetry. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Define line symmetry.
(ii) Find the lines of symmetry of a square, an equilateral triangle and a regular pentagon.
(iii) Draw them. 5

OR

(i) Define rotational symmetry.
(ii) Find the order and angle for a square, an equilateral triangle and a regular hexagon.
(iii) Which has the largest order? 5

20. (i) Classify the letters A, B, E, H, N, S, Z, O by line symmetry.
(ii) Which have rotational symmetry? 5

OR

Case study: A rangoli design is a regular hexagon with six petals. 5
(i) How many lines of symmetry?
(ii) Order of rotation?
(iii) Smallest angle?
(iv) Is it symmetrical about the centre?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND